

DRAFT FOR REGISTERED INDIAN PATENT PROFESSIONAL REVIEW — NOT FILED

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FORM 2 (proposed) | THE PATENTS ACT, 1970 | THE PATENTS RULES, 2003

COMPLETE SPECIFICATION
PROPOSED UNDER SECTION 10 — FOR AGENT REVIEW
A SYSTEM AND METHOD FOR ADAPTIVE CHILD DEVELOPMENT ROUTINE
GENERATION USING CONTEXT-AWARE ENVIRONMENTAL AND CAREGIVER-
ORIENTED COMPUTATIONAL PROCESSING

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Priority-boundary legend (used throughout)

P0 — FILED / PRIORITY-SUPPORTED disclosed in the 10 May 2026 Form 2 (37-page provisional PDF supplied by the applicant).

P1 — REASONABLY DERIVABLE / EMBODIMENT arguably within the filed written description; agent must confirm before relying for priority.

P2 — POST-FILING IMPLEMENTATION present in current production (often June 2026 engine freeze or later) and *not* clearly in Form 2 — **not** used as a priority limitation in the proposed claims.

P3 — FUTURE / SPECULATIVE wearable, voice-assistant generation, offline/local inference, smart-home, biometric, travel-mode, etc., as contemplated in Form 2 §6.10.

Only **P0/P1** material is used in the core Section 10 description intended to rely on the 10 May 2026 priority date. **P2/P3** appears only in labelled counsel sections or as alternative embodiments already present in Form 2 §6.10.

0. Counsel cover — what was actually filed versus this draft

The 10 May 2026 filing is a **provisional specification under Section 9**. The uploaded Form 2 is a 37-page print of an HTML package titled “AmyNest — Indian Provisional Patent Application Package”, generated 08/05/2026, 21:43 from file:///Users/macbook/Downloads/amynest_patent_package 2.html. Accompanying drawings are Figures 1–4 of that same print (drawing PDF: 4 sheets).

Form 2 itself describes a **hybrid** architecture: a generative language-model path *and* a deterministic rule-based path, with deterministic correction applied especially to probabilistic outputs, and (in Figure 3) “applied to both paths”. Current production (certified June 2026, after the filing date) uses a **rules-first** HTTP path as the primary generator, with an optional language-model path that is then passed through the same deterministic processing and may fall back to rules. That later primacy is **P2**. This Complete Specification therefore:

- 1. preserves the filed hybrid disclosure honestly;
- 2. does not recast the 10 May invention as “rules-first only”;
- 3. does not bury the environmental-synchronisation mechanism under generic parenting-AI language;
- 4. does not insert dinner-gap minutes, AQI caps, country-profile tables, meal-bank defaults, named intelligence-pipeline order, or HTTP 422 refuse into the priority description or proposed priority claims.

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1. Field of the Invention

P0 The present invention relates to computer-implemented child-development assistance systems. More particularly, it relates to the adaptive generation of child routines through hybrid processing comprising (i) deterministic rule-based orchestration and (ii) optional generative artificial-intelligence inference, together with environmental-condition integration, caregiver-adaptive computational transformation, asynchronous state coordination, and deterministic constraint/correction processing of generated routine artefacts.

Operative client execution environments disclosed for making and using the invention include a web-browser frontend execution environment and a native mobile frontend execution environment. Wearable, voice-assistant, and offline/local-inference environments are disclosed in the provisional as both multi-platform language and as future-proof embodiments (Form 2 §§6.1, 6.9, 6.10). In this Complete Specification they are treated as **alternative / contemplated embodiments** **P3** unless and until a patent professional directs otherwise. Cross-platform parity is claimed for the ambient-condition classification function and the deterministic constraint/correction infrastructure to the extent those modules are present on a given client.

2. Background of the Invention

P0 The management of a child’s daily routine is a computationally and cognitively demanding orchestration problem. Existing digital tools — general-purpose calendar applications, static parenting schedule templates, and isolated rule-based scheduling utilities — exhibit at least the following technical limitations (Form 2 §2):

1. **Static environmental handling.** Outdoor suitability is often entered manually. Omission yields a fixed default inconsistent with actual ambient conditions.
2. **Asynchronous environmental state conflicts.** Systems that fetch environmental data typically initiate a fresh network query on each invocation, producing duplicate device-permission prompts,

redundant computational requests, increased energy consumption, and concurrent asynchronous execution conflicts when multiple generation triggers fire in rapid succession.

3. **User-preference override risk.** A late-resolving environmental detection can retroactively overwrite an explicit user choice made during the detection window, producing user-interface synchronisation latency and loss of caregiver control.
4. **Probabilistic generation errors.** Outputs of a language-model inference engine may include developmentally inappropriate activities, activities during mandatory school hours, non-authentic dish names, incorrect dietary content, or repeated meal entries, unless a deterministic constraint/correction layer is applied.
5. **Excessive dependence on static input** without machine-controlled contextual transformation when environmental, caregiver, or developmental context changes.
6. **Caregiver-oblivious instructions** that do not adapt to the identity and capability profile of the active caregiver.
7. **Generic meal recommendations** that ignore ingredient inventory, regional cuisine, dietary restrictions, or allergy parameters.
8. **Static activity ordering** that does not adapt to a historically observed energy profile.

There is a need for a computational orchestration system and concurrent-safe scheduling architecture that (a) integrates real-time environmental data prior to schedule generation while preserving an explicit user choice; (b) suppresses duplicate network operations through shared in-flight retrieval state; (c) applies deterministic constraint/correction processing to generated artefacts, including probabilistic language-model outputs; (d) adapts instructions to the active caregiver; (e) produces region-aware, ingredient-aware, allergy-constrained meal options; and (f) may heuristically transform activity ordering according to observed energy profiles.

This background describes technical problems. It is not a prior-art search and does not assert that no prior art exists.

3. Objects of the Invention

It is an object of the invention to provide one or more of the following (Form 2 §3), without any single object being essential unless recited in a claim:

1. a silent, non-blocking, pre-generation ambient-condition classification module that derives an outdoor-suitability classification from geolocation and environmental data;
2. concurrent-safe execution management such that multiple simultaneous invocations share a single in-flight retrieval, suppressing duplicate network operations and device permission prompts;
3. an explicit user-preference preservation state flag that prevents a late-resolving environmental detection from overriding a preference set by the user during the asynchronous resolution window;
4. deterministic constraint/correction processing that constrains, anchors, and corrects generated routine artefacts, including probabilistic outputs of a language-model inference engine;

5. classification of children into developmental age bands and application of age-appropriate activity, meal, and sleep parameters;
6. caregiver-adaptive transformation of generated instructions according to the identity and capability of the active caregiver;
7. a meal-option enrichment mechanism producing region-specific, ingredient-aware, allergy-constrained meal options;
8. schedule-consistency enforcement preventing activity placement during mandatory school hours and enforcing temporal anchoring to the child's wake time;
9. minimisation of concurrent asynchronous execution conflicts and reduction of user-interface synchronisation latency through direct payload injection prior to a UI synchronisation cycle;
10. a multi-platform execution architecture in which the classification function and constraint/correction infrastructure may be realised on web and native-mobile clients, with further platforms as alternative embodiments.

4. Summary of the Invention

The invention is summarised in complementary aspects corresponding to Form 2 §4. Independent claim families are proposed in Section 7 for agent review; this summary does not itself define claim scope.

4.1 Computational orchestration system

P0 According to a first aspect, a computer-implemented computational orchestration system for adaptive generation of a child development routine comprises: an ambient-condition classification module; an age-band classification engine; a hybrid generation engine comprising a generative artificial-intelligence module and a deterministic rule-based path; deterministic constraint/correction processing; a caregiver-adaptive instruction transformation module; an environmental-state-driven activity substitution module; a meal-option enrichment engine; and an energy-profile-driven re-ordering module.

The hybrid generation engine is disclosed as providing **two paths**. The language-model inference engine is an interchangeable external component; no specific vendor or model name is a limitation of the invention. Deterministic rule-based processing and deterministic constraint/correction processing are **not** themselves “artificial intelligence” or “machine learning” as those terms are used herein.

4.2 Environmental-state synchronisation method (core technical mechanism)

P0 According to a second aspect, a computer-implemented method for environmental-state synchronisation prior to a routine-generation event comprises: evaluating an explicit user-preference preservation state flag; if not set, initiating a shared in-flight retrieval; mapping a returned environmental dataset to a categorical outdoor-suitability classification; threading said classification directly into the generation payload prior to a user-interface synchronisation cycle; and discarding a late-resolving classification if the preference flag has transitioned to a touched state during the asynchronous resolution window.

4.3 Concurrent-safe scheduling architecture

P0 According to a third aspect, multiple simultaneous invocations of a routine-generation trigger share a single in-flight environmental retrieval through a registry of pending asynchronous operations, thereby reducing redundant computational requests and minimising concurrent asynchronous execution conflicts.

4.4 Deterministic constraint/correction processing

P0 According to a fourth aspect, deterministic constraint/correction processing is operative at least on the output of the generative artificial-intelligence module, enforcing temporal anchoring, school-block exclusion, allergy filtering, cuisine authenticity validation, and cross-slot deduplication. Form 2 Figure 3 further discloses that a deterministic correction layer is “applied to both paths; AI path receives full chain”. Presenting constraint processing as applicable to rule-based artefacts as well as language-model artefacts is therefore **P0** as to the figure/body, even though several independent claims of the provisional recited correction “of probabilistic outputs” specifically. Agent direction is requested on whether independent Family C should remain LLM-output-specific (closest to provisional Claim 14) or be broadened to “generated routine artefacts from either path” (supported by Figure 3).

4.5 Execution environments

P0 / P3 According to a fifth aspect as filed, the invention was described as realised across web, native mobile, wearable, voice-assistant, and offline/local-inference environments. This Complete Specification treats web and native mobile as the operative enabling disclosure for making and using the invention, and treats wearable, voice-assistant, offline/local-inference, smart-home, biometric, multi-child, travel-mode, and educational-recommendation integrations as alternative or future embodiments (Form 2 §6.10), unless counsel elects the broader filed present-tense language.

FILED versus CURRENT PRODUCTION (do not merge timelines). Filed (10 May 2026): hybrid AI path + rule-based path; deterministic correction of probabilistic outputs; meal enrichment Phase B as a constrained language-model call producing a predetermined number of dish names per slot; platforms listed in present tense including wearable/voice/offline.

Current production (post-June 2026 certification): primary P0ST /routines/generate = resolveRoutineGenerationInputs → generateRuleBasedRoutine → runRoutineIntelligencePipeline → repairDinnerAnchor; optional P0ST /routines/generate-ai = language-model draft → same deterministic pipeline → safety handling → rules fallback. Meal banks are the default implemented meal mechanism. Dinner-gap 60/90/120 minutes, AQI outdoor caps, and named country-profile tables are post-filing implementations. See Annex A. None of those P2 details is used below as if it were disclosed on 10 May 2026.

5. Brief Description of Drawings

The accompanying drawings illustrate non-limiting embodiments. Figures 1–4 correspond in subject-matter to the drawings filed with the provisional (drawing PDF sheets 1–4). Figures 5–7 expand mechanisms already described in Form 2 §§6.2, 6.4.3 and 6.5; they do not introduce P2 production features.

1. **Figure 1** — Computational orchestration architecture: client execution environments, asynchronous execution layer, synchronisation / API gateway, generation modules, deterministic constraint/correction layer, and external service abstractions. Wearable, voice-assistant, and offline/local-inference blocks are shown dashed as alternative embodiments (reconciling Form 2 §6.1 present-tense list with §6.10 future-proof list).
2. **Figure 2** — Concurrent-safe environmental detection flow: explicit-preference preservation, in-flight retrieval sharing, duplicate-request suppression, classification mapping, late-resolution discard, and direct payload injection.
3. **Figure 3** — Hybrid generation pipeline: a path selector feeding a deterministic rule-based path and a generative language-model path in the alternative, both entering deterministic constraint/correction processing. This figure is redrawn so it does not imply that language-model generation is the sole or mandatory path. It also does not depict the later production “rules-first certified” order as the original invention.
4. **Figure 4** — Meal-option enrichment engine as filed: Phase A slot anchoring; Phase B constrained language-model option generation; ingredient interpolation; allergy/cuisine/dedup processing.
5. **Figure 5** — Deterministic environmental-state-driven activity substitution (Form 2 §6.5).
6. **Figure 6** — Late-resolution preference-preservation state machine (Form 2 §§6.2.1–6.2.3).
7. **Figure 7** — Deterministic constraint/correction processing chain disclosed in Form 2 §6.4.3 (temporal anchoring, school-block exclusion, anti-repetition, allergy/diet filtering, cuisine validation, cross-slot deduplication).

Filed Figure 2 caption routed the payload “to the Deterministic Correction Layer (AI hallucination correction path)”. The body and Figure 3 show generation occurring after environmental detection, with correction thereafter, including on the rule-based path. Figure 2 in this draft follows the body: detection → payload → generation path selector → constraint/correction. That is a consistency correction, not added technical matter. Agent should confirm.

6. Detailed Description of the Invention

Specific service names, model names, framework names, and source-function names appearing herein are illustrative implementation choices. The invention is not limited to any particular meteorological provider, language-model provider, speech-synthesis provider, operating system, or application framework. Where a source-file name is cited, it is *implementation evidence as of the date of this draft* and is not, by itself, a 10 May 2026 disclosure.

6.1 Overall Computational Architecture

P0 Referring to Figure 1, the invention is realised as a computational orchestration architecture comprising one or more client execution environments (100) and a server-side orchestration engine (300).

In an operative embodiment, client environments include a web-browser frontend execution environment (110) comprising an ambient-condition classification module (200), an asynchronous state-management layer having a synchronously readable reference structure (211) and a deferred component-state structure

(212), and a standard browser geolocation interface (230); and a native mobile frontend execution environment (120) in which geolocation is obtained through a native location module abstracting operating-system location services, and in which the same classification and payload-injection behaviour is provided.

Alternative embodiments (130, 140, 150), corresponding to Form 2 §§6.1 and 6.10, include a wearable computing device, a voice-assistant interface, and a local-inference embodiment in which selected modules execute on the client device without server connectivity. These are not required to make or use the operative embodiment.

The server-side orchestration engine (300) exposes a typed application-programming-interface gateway and houses: an age-band classification engine (310); a hybrid generation engine (320) comprising a generative artificial-intelligence module (321) and a deterministic rule-based path (322); deterministic constraint/correction processing (330); a meal-option enrichment engine (340); an energy-profile re-ordering module (350); a caregiver-adaptive instruction transformation module (360); and an environmental-state-driven activity substitution module (370). A persistent data layer (400) stores child profile data, observed behaviour analytics, and historical schedule artefacts.

External service abstractions include a remote meteorological data service (600) providing categorical and numerical environmental parameters; a language-model inference engine (500) providing probabilistic generation outputs, interchangeable among compatible providers; and, optionally, a speech-synthesis service (510) providing audio rendering of textual content. The architecture is service-agnostic.

Client-side state management is provided by an asynchronous state-management layer comprising a synchronously readable reference structure (211) and a deferred component-state structure (212), the cooperative use of which is described in §6.2.

Information flow, in conceptual order consistent with Form 2 Figures 2–3:

context inputs → environmental / user-preference synchronisation → age / caregiver / schedule / context processing → deterministic rule-based construction and/or optional probabilistic generation → deterministic constraint/consistency processing → environmental transformation → caregiver adaptation → meal enrichment → final routine artefact.

That order is a teaching aid. Form 2 does not require every module in every embodiment. The environmental-synchronisation mechanism of §6.2 may be practised with either generation path of §6.4.

6.2 Environmental Detection and Synchronization

P0 A core inventive concept, and the strongest clearly enabled concurrent-processing mechanism in the 10 May 2026 disclosure, is the ambient-condition classification module (200), implemented as an asynchronous helper invoked by routine-generation trigger handlers immediately prior to the generation request. The module performs concurrent-safe environmental detection orchestration, explicit user-preference preservation, classification mapping, and direct payload injection (Form 2 §6.2; Figure 2; proposed Claims 1 and 5).

6.2.1 Preference Preservation

The module first evaluates an explicit user-preference preservation state flag implemented as a pair of cooperating data structures within the asynchronous state-management layer: a synchronously readable reference structure (211) and a deferred component-state structure (212). Both data structures are transitioned to a touched state when the user explicitly engages with an environmental selection control.

When either data structure indicates a touched state at invocation time, the module returns the user-selected value immediately without initiating any environmental query. This implements a respect-the-explicit-user-choice invariant and prevents user-preference override.

The cooperative use of a synchronously readable reference together with a deferred component-state structure addresses a known asynchronous-state-coordination problem: the deferred component-state structure may not yet reflect the latest committed value inside an asynchronous closure (a stale-closure condition), whereas the synchronously readable reference structure is authoritative within the same execution turn. The invention thereby provides explicit user-preference preservation in the presence of asynchronous state coordination (Form 2 §6.2.1; Claim 13).

A platform-agnostic helper atomically updates both the synchronously readable reference and the deferred component-state structures so that both sentinel values are updated atomically from the calling context's perspective (Form 2 §6.9).

Implementation evidence (not a filing-date fact): in a web client this pair corresponds to a React state flag `weatherTouched` and a ref `weatherTouchedRef` in the routine-generation page.

6.2.2 Shared In-Flight Retrieval

A registry of pending asynchronous operations (220) holds at most one active environmental detection operation. When the module is first invoked in an untouched state, it assigns a new detection operation to the registry and may activate a user-interface indicator. Upon resolution or rejection, the registry entry is cleared and the indicator is deactivated.

If a second invocation occurs before the first detection resolves (for example, where the user triggers the generation control in rapid succession or where multiple generation-trigger pathways fire concurrently), the second invocation finds the registry populated and awaits the already-pending operation rather than initiating a second environmental query. Device permission dialogue and the upstream environmental query are each invoked at most once per generation session, providing duplicate-request suppression and reducing redundant computational requests (Form 2 §6.2.2; Claim 5A).

Implementation evidence: a module-level in-flight Promise `weatherDetectInFlightRef`. A server-side generate semaphore is a different mechanism and is not this registry.

6.2.3 Late-Resolution Race Handling

After the in-flight detection operation resolves, the module re-evaluates the synchronously readable reference structure of the explicit user-preference preservation flag before applying the detected classification. If the user has engaged the environmental selection control during the asynchronous

resolution window, the flag is now in the touched state and the detected classification is silently discarded; the user-selected value is returned instead. The module therefore cannot retroactively override an explicit user preference that arose after the generation trigger fired (Form 2 §6.2.3; Figure 6).

6.2.4 Environmental Classification

On a web-browser execution environment, the module obtains device coordinates through a standard browser geolocation interface with a configurable timeout and a position cache. On a native mobile execution environment, the module obtains coordinates through a native location module that abstracts operating-system location services. In alternative embodiments, coordinates may be obtained from an on-device wearable sensor, a paired companion device, or a voice-assistant location-context provider (Form 2 §6.2.4).

The obtained coordinates are submitted to the remote meteorological data service (600) to retrieve environmental parameters including a standard weather-condition code, an ambient temperature value, a precipitation value, and a wind-speed value. A deterministic classification mapping function derives one of a fixed set of categorical outdoor-suitability classifications:

Condition (any one sufficient)	Classification
Severe weather codes (thunderstorm, snow, rain showers, sustained rain); precipitation ≥ 2 mm; wind speed ≥ 40 km/h; temperature ≥ 40°C or ≤ 0°C	“unsuitable” for outdoor activity
Moderate weather codes (drizzle, fog); precipitation > 0 mm; temperature ≥ 35°C or ≤ 5°C; wind speed ≥ 25 km/h	“limited” — caution advised
All other conditions (clear, mainly clear, partly cloudy, overcast, calm)	“suitable” for outdoor activity

The derived classification is threaded directly as a parameter through the generation call chain and inserted into the generation payload before a user-interface synchronisation cycle is permitted to commit any related state (direct payload injection (260)). This guarantees that the freshly derived classification reaches the orchestration engine even where the frontend has not yet committed corresponding state, reducing user-interface synchronisation latency (Form 2 §6.2.4; Claim 5B).

Where a routine-generation flow is suspended pending a confirmation dialogue (for example, a wake-time confirmation), the derived classification is captured in a pending-action closure and re-injected upon dialogue confirmation, providing continuity of the derived classification through asynchronous user-interface interruptions (Form 2 §6.2.4; Claim 3).

Implementation evidence: UI tokens may be labelled yes / limited / no while the computational classification remains suitable / limited / unsuitable. That mapping is an interface embodiment of the filed three-way classification P1.

Air-quality index (AQI) outdoor-duration caps (0 / 15 / 20 minutes) exist in current production (maxOutdoorMinutesFromAqi) and were **not** disclosed in Form 2. They are P2 and are not part of this §6.2

6.2.5 Failure and Fallback

P0 The classification module catches exceptions including permission denial, network error, timeout, and abort signal, returning a null classification without rethrowing. On a null return the routine-generation flow proceeds with the existing user-selected or default value. The user is not interrupted by a detection failure; the orchestration is non-blocking (Form 2 §6.2.5; Claim 2).

6.3 Developmental Age Classification

P0 The invention classifies each child into one of a plurality of mutually exclusive developmental age bands based on the child’s age in months at the time of routine generation (Form 2 §6.3):

Age band	Age range (as filed)	Key parameters (as filed)
Infant	0–11 months	Caregiver-led sensory activities; 14–17 hour sleep; wake-and-feed windows; tummy time; no independent activities
Toddler	12–35 months	Movement, sensory play, imitation, early language; 90-minute afternoon nap anchor; playschool simplified logic
Preschool	36–59 months	Imagination, early literacy, sensory play; kitchen helper activities; rest block (not full nap)
Early School	60–119 months	Homework blocks; school-block enforcement; study skills; independence goals
Pre-Teen	120–180 months	Sustained study technique; identity and wellness activities; abstract reasoning tasks

The age band is computed within the orchestration engine from the child’s date of birth stored in the persistent data layer. Age-band-specific guidance strings are injected into the prompt supplied to the language-model inference engine to constrain vocabulary and category of generated activities. A date-seeded pseudo-random shuffle function, seeded with at least the generation date and a child identifier, is applied to activity pools in the rule-based path, providing day-to-day variation while remaining deterministic for a given date (Form 2 §6.3).

P1 — COUNSEL / TECHNICAL RECONCILIATION REQUIRED (age classifiers). The filed table above is the priority age-band disclosure. Current production contains additional classifiers that do *not* match this table: (i) kidschedule getAgeGroup — infant <12 / toddler <36 / preschool <60 / early_school <120 / pre_teen (aligned with the filed table); (ii) feeding getAgeGroup in routine-age-feeding.ts — infant_0_6 / infant_6_12 / toddler / child; (iii) safety-library classifyAgeBand — infant <18 / toddler <36 / preschool <60 / school <132 / tween. Safety validateRoutine is a separate advisory route, not the generate choke. Do not merge these thresholds in a claim. Do not present the safety-library cut-points as the 10 May invention.

6.4 Routine Generation Architecture

P0 The orchestration engine provides two routine-generation paths accessible via distinct interface endpoints (Form 2 §6.4; Figure 3).

6.4.1 Deterministic Rule-Based Generation

The rule-based path is a deterministic generation engine that selects activities from age-band-specific template pools. Inputs include the child age band, wake-up time, sleep time, school start/end times, mood, caregiver, environmental classification, parent goals, and ingredient inventory. The engine assembles a routine by sequentially filling a time cursor from wake to sleep, inserting mandatory anchor blocks (meals, hygiene, school where applicable, nap for toddlers, homework for school-age) and filling remaining slots from seeded-shuffled activity pools. The caregiver identity parameter modulates a bonding-density attribute — the number and type of caregiver–child joint activities inserted (Form 2 §6.4.1).

Form 2 describes this path as “zero-cost”. That phrase is an implementation characterisation, not a claim limitation.

6.4.2 Probabilistic / AI Generation

The generative path uses a language-model inference engine through a structured prompt. The prompt incorporates the child profile, date, age-band guidance, meal guidance, caregiver prompt, parent goals, mood context, observed energy profile, prior-day activity and meal categories, and the derived environmental classification (Form 2 §6.4.2). The language-model component is optional in the sense that the rule-based path of §6.4.1 is a complete generation embodiment without it. No vendor name (including any particular hosted model) is essential.

Language-model inference is probabilistic. The specification does not teach that a language model guarantees safety, developmental appropriateness, dietary correctness, or schedule consistency. Those properties, where obtained, are the result of deterministic constraint/correction processing and of the rule-based path.

6.4.3 Deterministic Constraint Processing

P0 Outputs of the language-model inference engine are post-processed through deterministic constraint/correction processing (330), which performs at least the following transformations (Form 2 §6.4.3; Figure 7):

- **Temporal anchoring:** a time-cascade utility re-anchors activity start times to the actual wake-up time, correcting temporal drift.
- **Schedule-consistency enforcement (school-block exclusion):** any activity item whose interval overlaps the child’s school start and end times is identified and rescheduled, preventing school-hour activities that are inconsistent with the stored school block.
- **Anti-repetition constraint:** prior-day meal categories and top activity categories are passed to the prompt and cross-checked in post-processing to enforce variety across consecutive days.

- **Allergy and dietary filtering:** items conflicting with the child’s allergy list or family dietary profile are removed and substituted from a deterministic safe-set.
- **Cuisine authenticity validation:** generated dish names are validated against a regional canonical-name registry; non-conforming names are rewritten or replaced.
- **Cross-slot deduplication:** the same dish name does not appear in more than one meal slot within the same daily routine.

Figure 3 of the provisional states that the deterministic correction layer is applied to both paths, with the AI path receiving the full chain. An embodiment therefore applies some or all of the above transformations to a rule-based artefact as well as to a language-model artefact.

The named production function sequence `runRoutineIntelligencePipeline` and the dinner-anchor repair `repairDinnerAnchor` with 60/90/120-minute gaps are P2. They may be argued as later implementations of “temporal anchoring” / “schedule-consistency” only with agent approval. They are not recited in the proposed priority claims. See Annex A.

6.4.4 Energy-Based Reordering

P0 Both paths may apply an energy-profile re-ordering step. The module reads the child’s historically observed energy profile (computed from stored routine-completion analytics) to identify a peak-focus window, a low-energy window, and a calm wind-down window. Cognitively demanding activities are shifted toward the peak-focus window; rest, sensory, and free-play activities toward the low-energy window; story time and bonding activities toward the calm window. Re-ordering is applied only when a minimum sample count of observations is available; otherwise the existing order is preserved (Form 2 §6.4.4).

Implementation evidence: a minimum sample count of three observations is used in current analytics code. The number “3” is not in Form 2 (which says “a minimum sample count”). Reciting “three” in a claim is P1 at most; safer to keep “a predetermined minimum sample count”.

This energy-profile processing is a deterministic heuristic over stored completion analytics. It is not a trained neural-network model and is not described as autonomous machine learning.

6.5 Environmental Activity Transformation

P0 Referring to Figure 5, the orchestration engine applies an environmental-state-driven substitution function (370) to generated routine items (Form 2 §6.5). Outdoor activities are identified by (i) an exact category match against a configurable outdoor category set, and (ii) a regular-expression test against a configurable outdoor keyword list (including, without limitation, park, playground, cycling, walk, garden, swim, run, jog, football, cricket, tennis, skating, fresh air).

For an “unsuitable” classification, outdoor items are substituted with pre-defined indoor equivalents from a deterministic lookup table, with replacement notes explaining the substitution. For a “limited” classification, outdoor items are retained but their duration is reduced and an indoor backup note is

appended. For a “suitable” classification, items are returned unchanged. The function is pure and non-mutating, providing referentially transparent behaviour suitable for concurrent-safe execution.

UAE-specific outdoor timing rules in current production are P2 and are not described here as priority subject-matter.

6.6 Caregiver-Adaptive Transformation

P0 After routine generation, a caregiver-adaptive instruction transformation module (360) rewrites routine-item instruction notes based on the active caregiver identity parameter (Form 2 §6.6). Disclosed profiles and policies:

Caregiver profile	Instruction adaptation policy (as filed)
First primary parental caregiver	Warm nurturing tone; meal-prep and bonding activities foregrounded
Second primary parental caregiver	Active play, confidence-building tasks; collaborative meal prep
Both primary parental caregivers	Additional family-bonding block (shared meal, joint outing, co-read)
Grandparent	Simple, low-energy activities; explicit step-by-step instructions; reduced demanding outdoor sports or complex cooking
Babysitter	Safe, structured, easy-to-supervise activities; pre-prepared meals; explicit safety/contact notes

Additionally, a contextual-prompt appender module appends caregiver-specific contextual prompts to the routine’s special-plan field prior to a call to the language-model inference engine. Combined pre- and post-generation transformation provides caregiver-adaptive instruction transformation.

Current production `simplifyForHandler` primarily prefixes activity titles (for example “Easy:” / “Step:”) rather than performing a general natural-language rewrite of all instruction notes. Bonding-density modulation in templates and caregiver strings in language-model prompts also exist. Proposed Claim 8 should not be broader than the written description plus what a skilled person would understand. Agent: consider narrowing “rewrite routine-item instruction notes” to “adapt activity presentation and/or instruction text according to caregiver profile”, or limit to the disclosed profiles.

6.7 Meal-Option Enrichment

P0 After the primary routine is generated, the orchestration engine may apply a two-phase meal-option enrichment engine (340) (Form 2 §6.7; Figure 4).

Phase A — Meal slot anchoring: ensure the routine contains correctly timed and labelled meal slots (Breakfast, Morning Snack, Lunch, Evening Snack, Dinner, Tiffin as applicable) consistent with the child’s age band and wake/sleep windows. Slot anchoring is also inherent in the rule-based path of §6.4.1, which inserts mandatory meal anchor blocks.

Phase B — Generative meal-option generation: a focused secondary call to the language-model inference engine produces a predetermined number of distinct meal-option names per meal slot. The enrichment prompt is parameterised by (a) family food style (vegetarian, non-vegetarian, vegan, Jain), (b) regional sub-cuisine, (c) parent-supplied ingredient inventory (treated strictly as ingredient names; instruction-like wording within the inventory is sanitised), (d) allergy list, and (e) the child’s age band (portion sizes and texture guidance).

Where an ingredient inventory is supplied, the enrichment prompt instructs the model to construct dish names by interpolating the supplied ingredients (for example, input “Paneer, Tomato” → output “Paneer paratha with tomato chutney”). The regional cuisine constraint governs cooking style; the ingredient list takes priority over regional bank suggestions. A deterministic deduplication step ensures the same dish name is not selected for more than one meal slot. The client presents enriched options as a selectable list.

Form 2 Claim 10 recites regional cuisine selected from a set comprising at least North Indian, South Indian, Bengali, Western, Asian, and Pan-Indian, with a region-specific description of canonical dishes incorporated in the prompt.

Meal banks as default production mechanism are P2 as a “default”. Form 2 already discloses deterministic template pools and meal anchors on the rule-based path (§6.4.1) P0. What was *not* disclosed is that deterministic meal banks are the primary production default and that “exactly four language-model options per slot” is not always the default. This Complete Specification therefore: (i) keeps Phase B as an optional generative enrichment embodiment as filed; (ii) does not claim “exactly four options” as essential; (iii) does not insert later meal-bank catalogues, country dinner windows, or sanitizer tables into the priority narrative. A dependent claim that meal names may be selected from deterministic template pools (without a language model) is P1 from §6.4.1 — see proposed Claim 15.

Jain / dietary logic — P1 claim-scope flag. Form 2 §6.7 and Claim 11 disclose Jain constraints as prompt constraints: exclude root vegetables in addition to meat, fish, and eggs, and exclude onion and garlic from suggested options. Figure 4 likewise shows “Jain → exclude meat · fish · eggs · root veg + exclude onion · garlic”. Current production encodes the same Jain root/onion/garlic rules in language-model prompt text (`artifacts/api-server/src/routes/routines.ts`). The deterministic meal-option sanitizer treats the Jain branch analogously to vegetarian for meat/fish/egg detection and does **not** implement a separate root-vegetable parser. **Do not claim a deterministic Jain root-vegetable enforcement engine on the 10 May priority date.** Claim 11 should remain prompt-constrained, or be omitted / rewritten with agent direction.

6.8 Engagement / Reward Processing

P0 Each routine item may be assigned reward points via a configurable weighting utility based on activity category: essential activities (hygiene, meals) receive a base weight; important activities (homework, study) receive a higher weight; play and creative activities receive the highest weight. Differential weighting supports balanced-day incentivisation and gamified progress tracking on the client interface (Form 2 §6.8). Reward points are an optional engagement layer, not a required limitation of the environmental-synchronisation or generation claims.

6.9 Multi-Platform Architecture

P0 The classification function and the deterministic constraint/correction infrastructure are disclosed as functionally identical across client execution environments, with platform-specific abstractions confined to the geolocation provider and the device-permission interface (Form 2 §6.9).

Operative enabling platforms for this Complete Specification: a web-browser frontend execution environment; and a native mobile frontend execution environment (which may host the same web implementation in a native shell).

Implementation evidence as of this draft (not a 10 May 2026 platform list): production clients are a web application, an iOS application delivering the web implementation via a Capacitor shell, and an Android application delivering the production site in a WebView. Those product facts may be used as enablement examples of “web-browser” and “native mobile” embodiments. They are not additional inventions.

Form 2 used present-tense language for wearable, voice-assistant, and offline/local-inference while also listing them under Future-Proof Embodiments (§6.10). This specification does not state that wearable generation, voice-assistant generation, or offline/local inference is an operative implemented embodiment. Proposed Claims 4 and 12 are narrowed to web and native mobile as the selected environments, which is within the filed “two or more ... selected from” list (provisional Claim 4).

6.10 Alternative and Future Embodiments

P3 Without limiting the scope of the claims as finally allowed, Form 2 §6.10 expressly contemplates additional embodiments. They are preserved here as alternative or future implementations, not as statements of current operation:

- Wearable-device integration, including biometric inputs (heart rate, activity, sleep stage) as additional generation parameters.
- Biometric sensor inputs to refine energy-profile windows (for example heart-rate variability, motion intensity).
- Smart-home integration in which the orchestration engine emits routine events to a controller (lighting, climate, audio).
- Voice-assistant integration presenting items audibly and accepting spoken status updates, with the preference-preservation flag set by spoken acknowledgement.
- Offline / local-device inference in which a reduced-footprint language model executes locally while preserving deterministic constraint/correction processing.
- Emotional-state inference from caregiver inputs, biometric inputs, or facial-expression analysis modulating bonding density and activity selection. Caregiver-entered mood as an input to generation is disclosed in §§6.4.1–6.4.2 **P0**; automated facial-expression analysis is **P3**.
- Multi-child orchestration with a shared-resource conflict-resolution layer.
- Vehicle / travel-mode adaptation substituting location-bound activities during detected travel intervals.
- Educational-recommendation integration ingesting learning-platform progress and biasing activity selection toward skill-gap-closing activities.

Dashed blocks in Figure 1 correspond to these alternative embodiments.

6.11 Technical Effects

The following technical effects are disclosed in Form 2 (abstract, §2, §3, §6.2, note on Section 3(k)) and are presented as engineering consequences of the architecture, not as a legal conclusion of patentability:

- reduction of redundant computational requests and suppression of duplicate network operations and duplicate device-permission prompts;
- minimisation of concurrent asynchronous execution conflicts through a single in-flight environmental retrieval;
- preservation of an explicit caregiver environmental preference in the presence of late-resolving asynchronous detection (stale-closure-safe dual-state flag);
- reduction of user-interface synchronisation latency by direct payload injection of the classification before a UI state-commit cycle;
- continuity of a derived classification through an asynchronous confirmation dialogue via a pending-action closure;
- non-blocking degradation: detection failure yields a null classification and generation proceeds on an existing or default value;
- deterministic schedule consistency: wake-time temporal anchoring and school-block exclusion applied by machine to generated artefacts;
- environmental-state-driven computational substitution of outdoor activities according to a three-way classification;
- deterministic post-processing of probabilistic language-model outputs (allergy/cuisine/dedup/anti-repetition);
- day-to-day variation with reproducibility via a date-and-child-identifier-seeded shuffle;
- referentially transparent (pure, non-mutating) environmental substitution suitable for concurrent-safe execution.

6.12 Section 3(k) Technical Character — for Agent Review

These technical features and effects are presented for patent-agent consideration in assessing Section 3(k) of the Patents Act, 1970 (computer program per se / algorithm / business method). **This draft does not state that the invention is patentable, that Section 3(k) is overcome, or that a practical result alone suffices.** No examination guideline is applied as a legal conclusion. Counsel must decide the final legal framing, including whether to emphasise the environmental-synchronisation family as the lead independent claim.

Features that a patent professional may consider as going to technical character of a computer-implemented invention, as disclosed:

1. **Asynchronous state coordination** using a synchronously readable reference that is authoritative inside an asynchronous closure together with a deferred component-state structure that drives rendering.

2. **Shared in-flight operation registry** holding at most one environmental detection, so concurrent triggers share one geolocation permission and one meteorological query.
3. **Duplicate network-operation suppression** and consequent reduction of device energy and radio use associated with repeated queries.
4. **Explicit preference preservation under asynchronous resolution**, including discard of a late result after the user has touched the control.
5. **Direct payload injection** of a derived classification into the generation call chain before UI state synchronisation, addressing stale-state delivery of environmental parameters to the server.
6. **Pending-action continuity** of that payload across an asynchronous confirmation dialogue.
7. **Machine-controlled temporal anchoring and school-block exclusion** applied as deterministic transformations of a generated artefact.
8. **Environmental-state-driven computational substitution** using category/keyword identification and a deterministic indoor-equivalent lookup table, as a pure function.
9. **Deterministic post-processing of probabilistic outputs** of an interchangeable language-model component, so correctness properties are not entrusted to the model.
10. **Concurrent-safe orchestration** of the above in a multi-trigger client.

If counsel elects to lead with Family A (environmental synchronisation), Families B–E may be pursued as dependent claims, additional independent claims, or later divisionals under Section 16, as already contemplated in the provisional's continuation/divisional note.

7. Proposed Claims

FOR REGISTERED PATENT PROFESSIONAL REVIEW ONLY. These claims are a technically supported working set. They are not legally final, not certified allowable, and not a novelty/inventive-step opinion. Priority support ratings: STRONG / MODERATE / WEAK / UNSUPPORTED. Full matrix: `patent/amynest_section10_claim_support_matrix.md`.

1. **(Independent — Family A — system)** A computer-implemented computational orchestration system for adaptive routine generation, comprising:
 - a. an explicit user-preference preservation state flag comprising a synchronously readable data structure and a deferred component-state data structure within an asynchronous state-management layer, both transitioned to a touched state upon explicit user engagement with an environmental selection control;
 - b. a registry of pending asynchronous operations configured to hold at most one active environmental detection operation, such that concurrent invocations of a routine-generation trigger share said operation, providing duplicate request suppression;
 - c. a geolocation interface module configured to obtain device coordinates through a platform-provided geolocation provider with a configurable timeout;
 - d. an environmental data interface module configured to query a remote meteorological data service using said coordinates and to receive at least one of: a standard weather-condition code, an ambient temperature value, a precipitation value, and a wind-speed value;

- e. an ambient-condition classification module configured to map the received environmental parameters to one of a fixed set of outdoor-suitability classifications;
- f. an explicit-preference preservation module configured to re-evaluate the synchronously readable data structure after the asynchronous detection operation resolves, and to discard the derived classification and return the user-selected value where the flag has transitioned to the touched state during the asynchronous resolution window; and
- g. a direct payload injection module configured to thread the derived or preserved classification as a parameter through the routine-generation call chain prior to a user-interface synchronisation cycle, thereby reducing user-interface synchronisation latency.

PRIORITY SUPPORT: STRONG — Form 2 §4 second/third aspects, §6.2, provisional Claim 1, Figure 2.

2. **(Dependent on 1)** The system of claim 1, wherein the geolocation interface module operates non-blockingly, catching exceptions including permission denial, network error, timeout, and abort signal, and returning a null classification on failure such that the routine-generation flow proceeds with an existing user-selected or default value. PRIORITY SUPPORT: STRONG — Form 2 §6.2.5, provisional Claim 2.
3. **(Dependent on 1)** The system of claim 1, wherein the routine-generation call chain comprises an asynchronous confirmation dialogue, and wherein the derived classification is captured in a pending-action closure prior to dialogue presentation and re-injected into the routine-generation payload upon dialogue confirmation. PRIORITY SUPPORT: STRONG — Form 2 §6.2.4 last paragraph, provisional Claim 3.
4. **(Dependent on 1 — platforms narrowed)** The system of claim 1, implemented across at least two client execution environments selected from a web-browser frontend execution environment and a native mobile frontend execution environment, the ambient-condition classification module being functionally identical across said execution environments. PRIORITY SUPPORT: STRONG for web + native mobile (within provisional Claim 4's "selected from" list). Wearable, voice-assistant, and offline/local-inference are not recited. Agent review: whether to restore the longer list as optional selected environments. NEW-MATTER RISK if wearable/voice/offline are claimed as *operative implemented* embodiments rather than alternatives.
5. **(Independent — Family A — method)** A computer-implemented method for environmental-state synchronisation in a routine-generation pipeline, comprising:
 - a. receiving a routine-generation trigger;
 - b. evaluating an explicit user-preference preservation state flag comprising a synchronously readable data structure and a deferred component-state data structure;
 - c. where either data structure indicates explicit user engagement, returning the user-selected value;
 - d. where neither data structure indicates explicit user engagement, querying a registry of pending asynchronous operations; awaiting an existing pending operation if present, or initiating a new environmental detection operation, registering said operation, and awaiting its resolution;
 - e. upon resolution, re-evaluating the synchronously readable data structure and discarding the derived classification where the flag has transitioned to the touched state during the asynchronous resolution window;

- f. where the derived classification is non-null and explicit user engagement has not occurred, returning the derived classification; and
- g. injecting the returned classification directly into a routine-generation payload prior to a user-interface synchronisation cycle.

PRIORITY SUPPORT: STRONG — Form 2 §4 second aspect, §6.2, provisional Claim 5. Antecedent “the derived classification” in (e) presupposes that step (d) produced a classification; agent may insert “if a classification is derived,” for clarity.

6. **(Dependent on 5)** The method of claim 5, wherein step (d) provides shared in-flight retrieval state such that two or more concurrent invocations of the method share a single environmental detection operation, thereby reducing redundant computational requests and minimising concurrent asynchronous execution conflicts. PRIORITY SUPPORT: STRONG — Form 2 §6.2.2, provisional Claim 5A.
7. **(Dependent on 5)** The method of claim 5, wherein step (g) bypasses the deferred component-state structure and transmits the classification directly into the generation payload, providing direct payload injection before a user-interface synchronisation cycle. PRIORITY SUPPORT: STRONG — Form 2 §6.2.4, provisional Claim 5B.
8. **(Independent — Family B — system)** A computer-implemented system for adaptive generation of a child development routine, comprising:
 - a. an age-band classification engine configured to assign a child to one of a plurality of developmental age bands, each band associated with a distinct set of activity vocabulary constraints, meal guidance, sleep parameters, and structural templates;
 - b. a date-seeded shuffle function configured to randomise activity pool ordering using a seed derived from at least the generation date and a child identifier, providing day-to-day variation while remaining reproducible;
 - c. a hybrid generation engine comprising a deterministic rule-based path operating on template activity pools and a generative artificial-intelligence module constrained by age-band guidance, the engine being configured to produce a routine artefact from the deterministic rule-based path, or from the generative artificial-intelligence module, or from both in combination; and
 - d. deterministic constraint/correction processing configured to operate on a routine artefact produced by the hybrid generation engine, providing at least: temporal anchoring of activity start times to a wake-time reference, and schedule-consistency enforcement excluding activity placement during mandatory school hours.

PRIORITY SUPPORT: MODERATE–STRONG. (a)(b) STRONG — Form 2 §6.3. (c) STRONG as hybrid comprising both modules — Form 2 §6.4; wording “or from both” is P1 clarification of two endpoints. (d) STRONG as to those two constraints — §6.4.3; allergy/cuisine/dedup moved to Claim 11 to avoid forcing LLM-only limitations into the independent claim.

Provisional Claim 6(d)–(e) recited correction operative on GAI outputs and anti-repetition into the GAI prompt; this redraft does not require a language-model output as a condition of (d), relying on Figure 3 “applied to both paths”. Agent must confirm that this is not an impermissible shift. If counsel prefers strict fidelity, restore “operative on outputs of the generative artificial-intelligence module” as in provisional Claim 6/14 and keep rule-based-only correction as a dependent.

9. **(Dependent on 8 — Family F)** The system of claim 8, further comprising an energy-profile re-ordering module configured to read a historically observed energy profile comprising at least a peak-focus window and a low-energy window, and, when a predetermined minimum sample count of observations is available, to reassign activity time slots such that cognitively demanding activities are placed within the peak-focus window and rest, sensory, or free-play activities within the low-energy window. PRIORITY SUPPORT: STRONG — Form 2 §6.4.4, provisional Claim 7. Do not recite sample count = 3 unless counsel accepts P1.
10. **(Dependent on 8 — Family D)** The system of claim 8, further comprising a caregiver-adaptive instruction transformation module configured to receive a caregiver identity parameter selected from a plurality of caregiver profiles and to adapt routine-item instruction text and/or activity selection with caregiver-appropriate language, complexity level, and safety emphasis corresponding to the selected profile. PRIORITY SUPPORT: MODERATE — Form 2 §6.6, provisional Claim 8. Narrowed from “rewrite ... notes” because production implements a more limited transformation. Agent review required.
11. **(Dependent on 8 — Family C / environmental transformation)** The system of claim 8, further comprising an environmental-state-driven activity substitution module configured to substitute outdoor activities with deterministic indoor equivalents responsive to an “unsuitable” classification, to reduce activity duration and append an indoor backup note responsive to a “limited” classification, and to pass items unchanged responsive to a “suitable” classification. PRIORITY SUPPORT: STRONG — Form 2 §6.5, provisional Claim 8A.
12. **(Dependent on 8)** The system of claim 8, wherein the deterministic constraint/correction processing further provides allergy filtering, cuisine authenticity validation against a regional canonical-name registry, and cross-slot deduplication of dish names. PRIORITY SUPPORT: STRONG — Form 2 §6.4.3. If independent Claim 8 is limited to GAI outputs, this dependent follows provisional Claim 6(d).
13. **(Dependent on 8)** The system of claim 8, further comprising an anti-repetition module configured to incorporate prior-day category frequencies into a prompt supplied to the generative artificial-intelligence module and/or to cross-check prior-day categories in post-processing. PRIORITY SUPPORT: STRONG for prompt incorporation — Form 2 §6.4.3 / provisional 6(e). “And/or post-processing” is P1 from the same paragraph (“passed to the prompt and cross-checked in post-processing”).
14. **(Independent — Family E — method)** A computer-implemented method for generating personalised meal options within a child development routine, comprising:
- anchoring meal slot times and labels within a generated routine based on the child’s age band and wake-up and sleep times;
 - receiving a parent-supplied ingredient inventory and at least one of: a regional cuisine parameter, a dietary restriction parameter, and an allergy list;
 - constructing a constrained natural-language prompt for a language-model inference engine instructing the engine to generate a predetermined number of distinct dish-name options per

- meal slot, using the supplied ingredients as primary components and constrained to the regional cuisine, dietary restriction, and allergy parameters;
- d. applying a cross-slot deduplication function ensuring no dish name appears in more than one meal slot within the same daily routine; and
- e. sanitising the parent-supplied ingredient inventory such that instruction-like wording therein is treated as ingredient names.

PRIORITY SUPPORT: STRONG as an optional enrichment method — Form 2 §6.7, provisional Claim 9. Do not present this as the only or default production meal mechanism. Deleted filed step “validating that the engine output contains the predetermined number of options per slot” may be restored if counsel wants fidelity; it is enablement, not essential. Do not claim “exactly four” options.

15. **(Dependent on 8 — template meals, not LLM-mandatory)** The system of claim 8, wherein the deterministic rule-based path inserts mandatory meal anchor blocks and selects meal names from age-band-specific template pools. PRIORITY SUPPORT: MODERATE / P1 — Form 2 §6.4.1 (mandatory meal anchors; template pools). Distinguishes template meals from Claim 14 Phase B. Not a meal-bank catalogue claim and not a P2 “banks are default” claim.
16. **(Dependent on 14)** The method of claim 14, wherein the regional cuisine parameter is selected from a set comprising at least North Indian, South Indian, Bengali, Western, Asian, and Pan-Indian, and wherein the natural-language prompt incorporates a region-specific description of canonical dishes, ingredients, and cooking techniques associated with the selected cuisine. PRIORITY SUPPORT: STRONG — Form 2 Claim 10, §6.7.
17. **(Dependent on 14 — Jain as prompt constraint)** The method of claim 14, wherein the dietary restriction parameter for a Jain dietary profile constrains the prompt to exclude meat, fish, and eggs, to exclude root vegetables, and to exclude onion and garlic from suggested options. PRIORITY SUPPORT: STRONG as prompt constraint — Form 2 Claim 11, Figure 4. UNSUPPORTED if amended to require a deterministic root-vegetable parser. Agent: keep “constrains the prompt”. P1 production gap: sanitizer does not enforce roots.
18. **(Independent — Family A coordination method)** A computer-implemented method of asynchronous state coordination in a routine-generation pipeline, comprising maintaining cooperatively a synchronously readable reference data structure and a deferred component-state data structure within an asynchronous state-management layer, such that the synchronously readable data structure is authoritative within an asynchronous closure for the purpose of explicit user-preference preservation while the deferred component-state data structure drives user-interface re-rendering, thereby reducing user-interface synchronisation latency and preventing override of explicit user choices by late-resolving asynchronous events. PRIORITY SUPPORT: STRONG — Form 2 §6.2.1, provisional Claim 13. May be objected to as abstract; consider depending from Claim 5.
19. **(Independent — Family C — method)** A computer-implemented method for deterministic correction of probabilistic outputs of a language-model inference engine in the context of child development routine generation, comprising:

- a. receiving a probabilistic routine output from the language-model inference engine;
- b. applying temporal anchoring to align activity start times with a wake-time reference;
- c. applying schedule-consistency enforcement to exclude activity placement during mandatory school hours;
- d. applying allergy filtering to remove items conflicting with stored allergy parameters and to substitute deterministic safe alternatives;
- e. applying cuisine authenticity validation against a regional canonical-name registry;
- f. applying cross-slot deduplication to remove repeated dish names within the same routine; and
- g. returning a corrected routine artefact for delivery to a client execution environment.

PRIORITY SUPPORT: STRONG — Form 2 §6.4.3, provisional Claim 14. This family remains LLM-output-specific as filed. Do not silently convert step (a) into “receiving a rule-based artefact” without agent approval (Figure 3 support exists but is a claim-shift).

20. **(Dependent on 19)** The method of claim 19, further comprising applying environmental-state-driven activity substitution responsive to a categorical outdoor-suitability classification, providing adaptive environmental transformation of the corrected routine artefact prior to delivery. PRIORITY SUPPORT: STRONG — Form 2 §6.5, provisional Claim 15.

NOT INCLUDED in the priority claim set (P2 / new-matter risk): dinner-to-bedtime gaps of 60 / 90 / 120 minutes; country-specific dinner or sleep windows; AQI outdoor caps of 0 / 15 / 20 minutes; HTTP 422 validation refuse as a claimed boundary; named “intelligence pipeline” pass order; family-intelligence trustScore; meal banks as the mandatory default; infant exclusive-phase skip under 6 months as a claimed pass-skip; language-model failure fallback to the rule-based path as an independent claim (fallback is a later production behaviour; Form 2 silent-failure is for environmental detection, not for LLM-to-rules fallback); vendor-specific models. See Annex A. **POST-FILING FEATURE — REMOVE FROM PRIORITY CLAIM OR OBTAIN PATENT-AGENT DIRECTION.**

Former provisional Claim 12 (multi-platform architecture requiring ambient-condition classification, deterministic correction, and meal-option enrichment to be functionally identical across web, native, wearable, voice, and offline) is *not* reproduced as an independent claim. It mixed Family A+C+E and recited unimplemented platforms in present tense. Enablement of functional identity is retained in §6.9. Agent may restore a narrowed two-platform claim if desired (overlaps Claim 4).

8. Abstract

Proposed Form 2 abstract. Target: not more than 150 words (Patents Rules, 2003, as commonly applied). Word count of the following paragraph: **139 words**.

A computational orchestration system generates adaptive child-development routines using environmental-state synchronisation and hybrid generation. Before generation, an ambient-condition classification module evaluates an explicit user-preference flag implemented as a synchronously readable reference and a deferred component-state structure. If untouched, concurrent invocations share one in-flight geolocation and meteorological retrieval. Returned weather-code, temperature, precipitation and wind parameters are mapped to a suitable, limited or unsuitable outdoor classification and injected into

the generation payload before a user-interface synchronisation cycle. A late-resolving detection is discarded if the flag becomes touched during the asynchronous window. A hybrid engine provides a deterministic rule-based path and an optional language-model path. Deterministic constraint processing applies temporal anchoring, school-block exclusion, allergy filtering, cuisine validation and meal-slot deduplication. Technical effects include duplicate-request suppression, preference preservation under asynchronous resolution, and reduced interface synchronisation latency. Language-model vendor identity is not a limitation.

9. Drawings

Figure 1 — Computational orchestration architecture

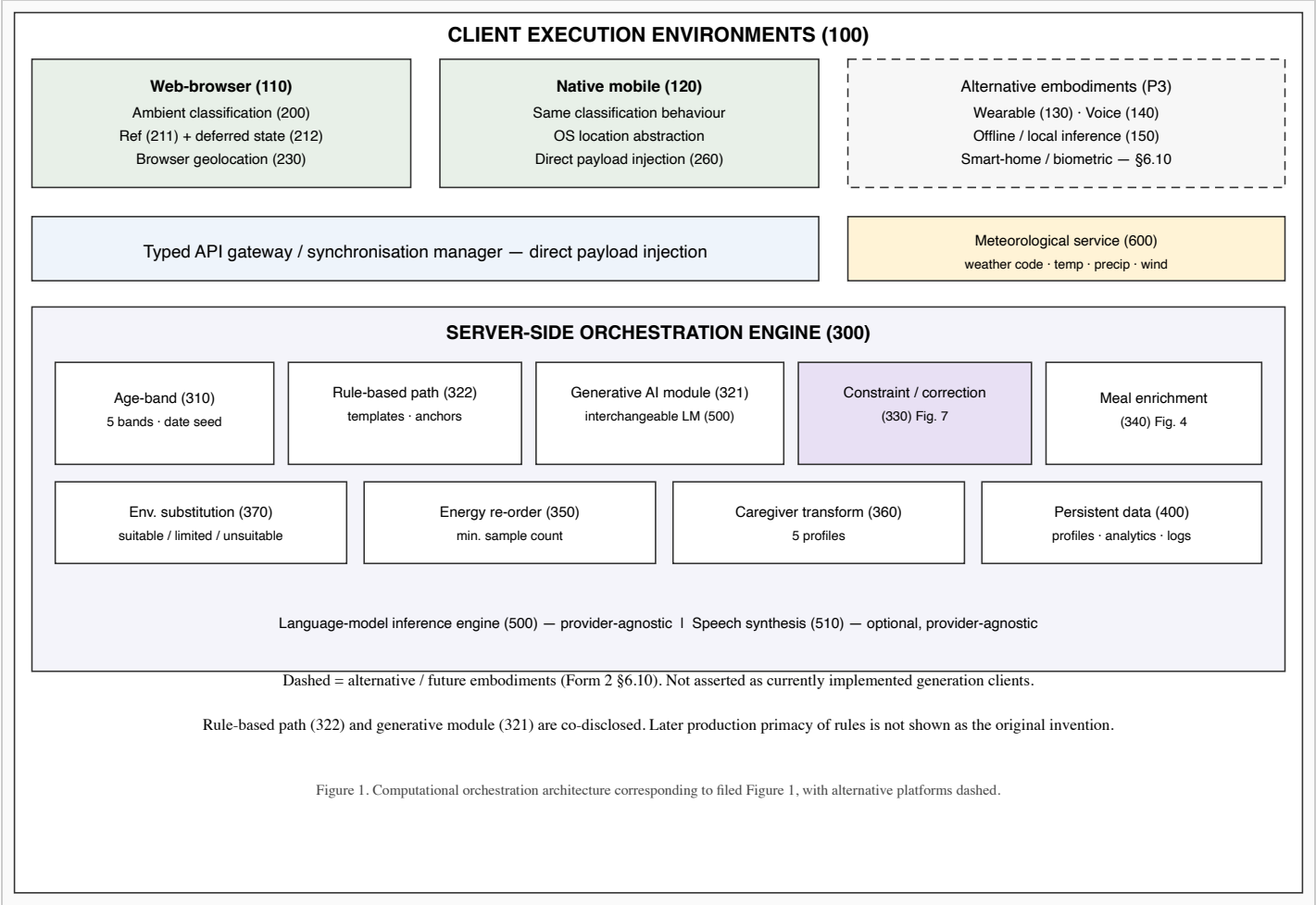


Figure 1 — Computational orchestration architecture: operative web and native-mobile clients, typed API gateway, server modules including deterministic constraint/correction processing, and provider-agnostic external services. Wearable, voice, and offline/local inference are dashed alternative embodiments.

Figure 2 — Concurrent-safe environmental detection flow

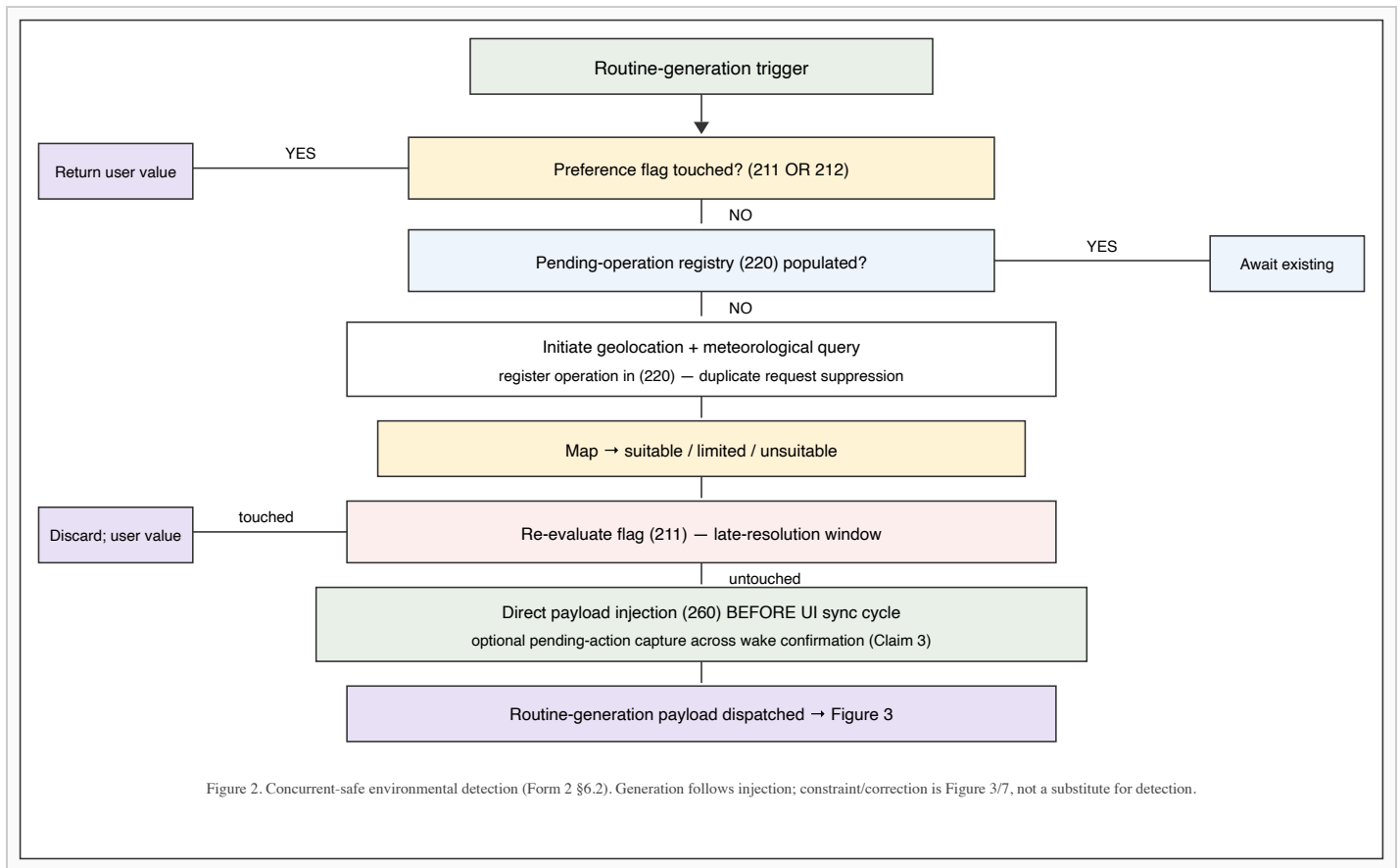


Figure 2 — Concurrent-safe environmental detection: dual-state preference flag, shared in-flight registry, three-way classification, late-resolution discard, and direct payload injection. Unlike the filed Figure 2 caption, this figure does not route detection directly into an “AI hallucination correction path”; that processing is Figure 7 after generation.

Figure 3 — Hybrid generation pipeline with deterministic constraint processing

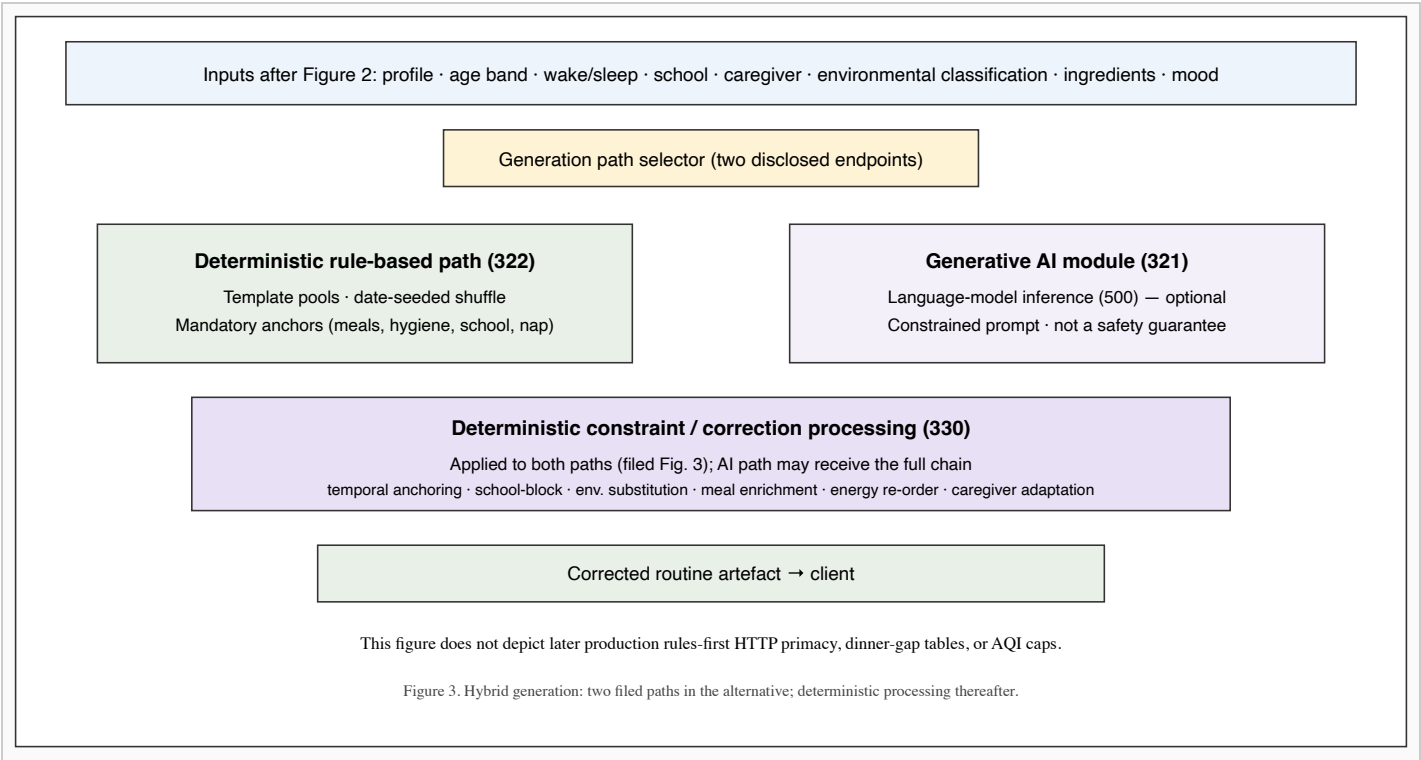


Figure 3 — Hybrid generation pipeline. Redrawn from filed Figure 3 so that the language-model path is not implied to be the sole invention, and so that later rules-first production order is not back-dated. Filed Figure 3 already showed a path selector and correction “applied to both paths”.

Figure 4 — Meal-option enrichment engine (as disclosed 10 May 2026)

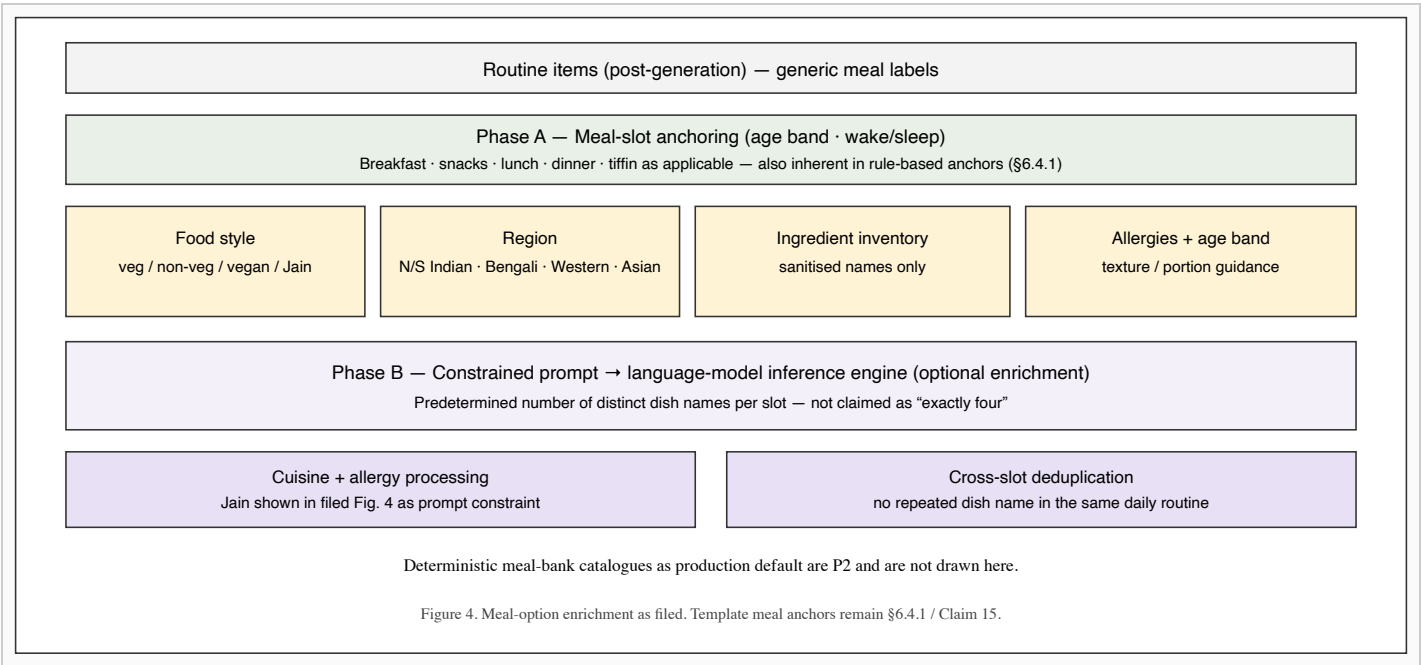


Figure 4 — Meal-option enrichment engine corresponding to filed Figure 4: Phase A anchoring and Phase B optional language-model options, with allergy/cuisine/dedup. Does not depict later meal-bank default architecture.

Figure 5 — Deterministic environmental activity transformation

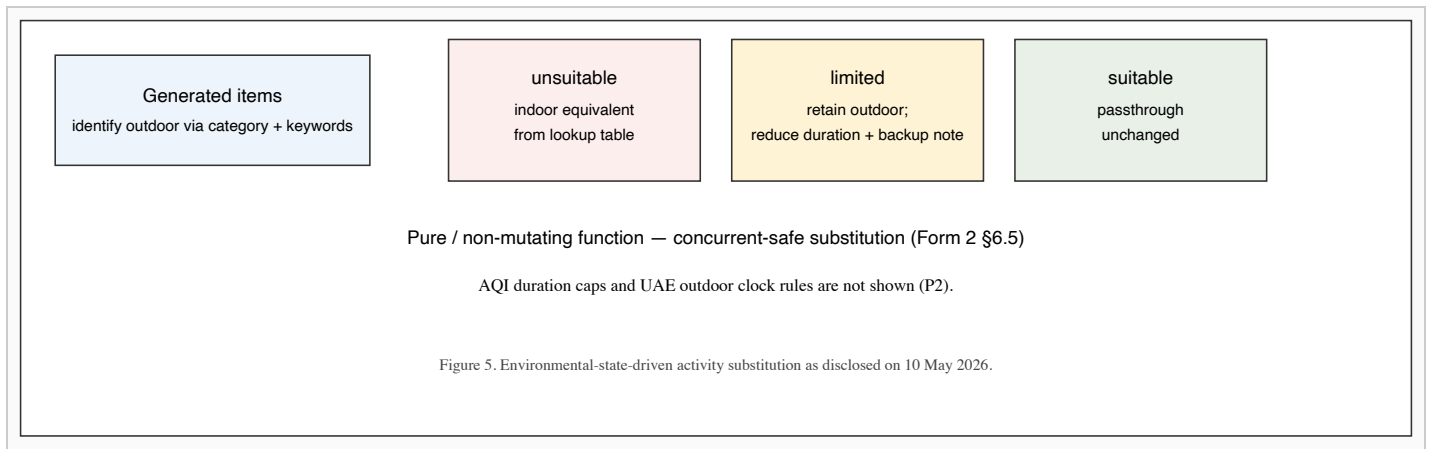


Figure 5 — Deterministic environmental transformation supported by Form 2 §6.5. No post-filing AQI or country-clock rules.

Figure 6 — Late-resolution preference-preservation state machine

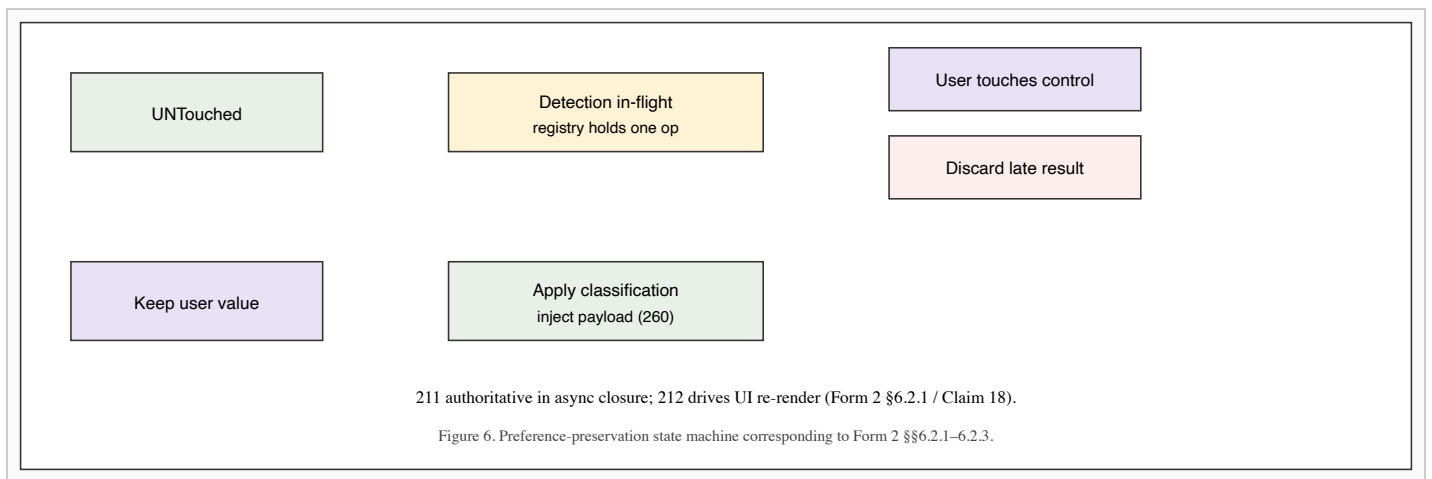


Figure 6 — State machine for explicit preference preservation and late-resolution discard. Supported by Form 2 §6.2; no new production mechanism.

Figure 7 — Deterministic constraint / correction processing

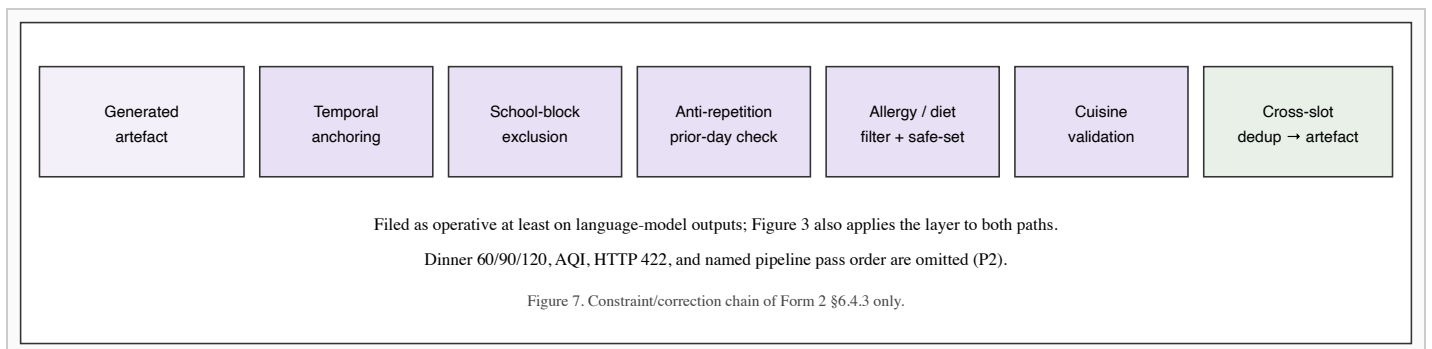


Figure 7 — Deterministic constraint/correction chain limited to mechanisms written in Form 2 §6.4.3. Not a drawing of the later production intelligence pipeline.

Annex A — Post-filing technical development / new-matter risk register

This annex is **not** part of the priority written description. It exists so that a registered patent professional can see what current production does without back-dating those details to 10 May 2026. Inclusion of any P2 item in a complete-specification claim or in the body as if originally disclosed requires agent direction (amendment strategy, extra examples as non-priority embodiments, and/or a later application).

Feature (production)	In Form 2 10 May 2026?	Broad filed concept?	Treatment in this draft	Claim / new-matter
Rules-first primary HTTP path; pipeline order; repairDinnerAnchor as last step	NO as primacy / named order	YES — rule-based path + correction on both paths	Counsel note only; Figure 3 keeps both paths	Do not claim certified order as priority
Dinner-bed gaps 60 / 90 / 120 min; country dinner windows	NO	WEAK — “temporal anchoring” / meal anchors	Annex only	P2 — remove from priority claims
AQI caps 0 / 15 / 20 min; UAE outdoor clock	NO	NO (weather table is WMO-style, not AQI)	Annex only	P2
Meal banks as default	NO as default	YES — template pools / meal anchors §6.4.1	Claim 15 P1; default status P2	Do not claim banks-as-default
LLM fallback to rules on generate-ai failure	NO (silent fail is weather-only)	REVIEW	Annex only	P2 as independent claim
HTTP 422 routine_validation_failed	NO	WEAK — correction “returns artefact”	Annex only	P2
Family intelligence / trustScore / learning weights	NO (analytics + energy profile only; educational rec is §6.10 future)	WEAK	Annex only; learning weights are prompt-level correlations, not a trained model	P2
Infant exclusive phase <6 months skip	NO (infant band is 0–11)	P1 at most	Annex only	P2 detail
Safety classifyAgeBand <18/<132	NO (filed table differs)	NO	§6.3 flag	Do not claim
Capacitor iOS / Android WebView product shells	NO by name	YES — web + native mobile	§6.9 implementation evidence	OK as examples of filed platforms

OpenAI / GPT-4o-mini	Not in this Form 2 (provider-agnostic language)	N/A	Not claimed	Do not add vendor limitation
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Annex B — Reference numerals

No.	Component
100	Client execution environments
110	Web-browser frontend execution environment
120	Native mobile frontend execution environment
130–150	Alternative wearable / voice / offline embodiments
200	Ambient-condition classification module
211	Synchronously readable reference structure
212	Deferred component-state structure
220	Registry of pending asynchronous operations
230	Geolocation interface
260	Direct payload injection
300	Server-side orchestration engine
310	Age-band classification engine
321	Generative artificial-intelligence module
322	Deterministic rule-based path
330	Deterministic constraint/correction processing
340	Meal-option enrichment engine
350	Energy-profile re-ordering module
360	Caregiver-adaptive transformation module
370	Environmental-state-driven activity substitution module
400	Persistent data layer
500	Language-model inference engine (interchangeable)
510	Speech-synthesis service (optional)
600	Remote meteorological data service

Public patent-status recommendation (do not publish automatically)

Product/118n/marketing strings were **not** edited in this task. Conservative wording for counsel approval only:

Indian Patent Application No. 202611059355 filed on 10 May 2026.

Do not use “patented”, “patent granted”, “patent approved”, or treat “Provisional patent filed” as equivalent to grant. Complete specification under Section 10 remains outstanding as of this draft (twelve-month clock from 10 May 2026).

End of technical Complete Specification draft.

Indian Patent Application No. 202611059355 | Not filed with IPO as this document | Not a legal opinion

Status for this package: TECHNICAL COMPLETE SPECIFICATION DRAFT READY FOR REGISTERED INDIAN PATENT
PROFESSIONAL REVIEW